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Global analysis of protein expression in muscle tissues of dermatomyositis/polymyosisits patients demonstrated an association between dysferlin and human leucocyte antigen A

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Abstract

Objectives. DM and PM are characterized by myofibre damage with inflammatory cell infiltration due to the strong expressions of MHC class I HLA-A and monocyte chemoattractant protein-1 (MCP-1). Dysferlin (DYSF) is a transmembrane glycoprotein that anchors in the sarcolemma of myofibres. DYSF mutation is closely associated with inherited myopathies. This study aimed to determine the role of DYSF in the development of DM/PM.

Methods. Mass spectrometry was performed in muscle tissues from DM/PM patients and controls. The DYSF levels in muscle tissue, peripheral blood cells and serum were detected by Western blotting, IF, flow cytometry or ELISA. Double IF and co-immunoprecipitation were used to investigate the relationship between DYSF and HLA-A.

Results. Mass spectrometry and bioinformatics analysis findings suggested the dysregulated proteins in DM/PM patients participated in common biological processes and pathways, such as the generation of precursor metabolites and energy. DYSF was upregulated in the muscle tissue and serum of DM/PM patients. DYSF was mainly expressed in myofibres and co-localized with HLA-A and MCP-1. DYSF and HLA-A expressions were elevated in myocytes and endothelial cells after being stimulated by patient serum and IFN-β. However, no direct interactions were found between DYSF and HLA-A by co-immunoprecipitation.

Conclusion. Our study revealed the dysregulated proteins involved in common and specific biological processes in DM/PM patient samples. DYSF is upregulated and exhibits a potential role along with that of HLA-A and MCP-1 in inflammatory cell infiltration and muscle damage during the development of DM/PM.

Key words: dermatomyositis, polymyositis, protein profiles, DYSF, HLA-A

Rheumatology key messages

- Mass spectrometry analysis revealed differentially and commonly altered proteins in DM and PM muscles.
- Dysferlin was upregulated in DM/PM and mainly located in the sarcolemma and cytoplasm of myofibres.
- Dysferlin was co-located with HLA-A and monocyte chemoattractant protein-1 and might be involved in the pathogenesis of DM/PM.

Introduction

Idiopathic inflammatory myopathies (IIMs) are a group of rare diseases characterized by proximal muscle weakness, elevated muscle enzyme levels and myositis-specific autoantibodies in the serum [1]. DM, PM, immune-mediated necrotizing myopathy and sporadic IBM (sIBM) are common subtypes of IIM in clinical practice, with high mortality rates due to severe muscle damage [2, 3]. Inflammatory cell infiltration into the muscle tissues that leads to myofibre necrosis and atrophy is a main characteristic of DM and PM [2]. $CD4^+$ T cell,

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BASIC SCIENCE

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macrophage and B cell infiltrations into the perimysial areas and perifascicular muscle fibre atrophy are key features of DM [1]. $CD8^+$ T cells, macrophages and $CD4^+$ T cells invading non-necrotic muscle fibres are visible in both PM and sIBM; the hallmark of the latter is amyloid deposit accumulation [4–6]. Recent studies revealed that inflammatory cell infiltration in DM and PM occurred due to the widespread appearance of MHC class I (MHC-I) molecules and elevated levels of chemoattractants such as monocyte chemoattractant protein-1 (MCP-1) in muscle fibres [1, 4, 7–12]. HLA-A, HLA-B and HLA-C are the major MHC-I molecules anchored within the membrane and play a central role in the immune system by presenting peptides to cytotoxic $CD8^+$ T cells [13]. MCP-1 displays chemotactic activity for monocytes and has been implicated in the pathogenesis of diseases characterized by monocyte infiltration, such as DM [14, 15]. In addition, in DM and PM, vessels in the muscle tissue are abnormally thickened and enlarged and may facilitate inflammatory cell trafficking [16–18]. However, the exact mechanisms underlying chemoattractant elevation and inflammatory cell infiltration in DM and PM remain to be fully elucidated.

Dysferlin (DYSF), an ~237 kDa type II transmembrane glycoprotein that belongs to the ferlin family [19], mainly anchors to the sarcolemma of the skeletal muscle fibres, monocytes and trophoblasts that interact with calpain 3 and mitsugumin 53 and participate in plasma membrane repair [20–24]. DYSF also plays a role in maintenance of the T-tubule system and in Ca^{2+} homeostasis, vesicle trafficking and membrane fusion in myoblasts [25–28]. DYSF dysregulation is closely associated with many inherited myopathies and autoimmune diseases [29–31]. Gene mutations of DYSF can lead to dysferlinopathy, which is characterized by two main phenotypes: limb-girdle muscular dystrophy type 2B, with primarily proximal weakness, and Miyoshi myopathy, with primarily distal weakness. DYSF mutation usually results in degenerating, regenerating or necrotizing muscle fibres in muscle sections that are similar to those seen in DM and PM but exhibit no DYSF staining in the sarcolemma or inflammatory cell infiltration [19, 29, 32]. DYSF abnormally co-aggregates with amyloid-β plaques in sIBM myofibres, and β-amyloid precursor protein overexpression in cultured myocytes increases the MHC-I expression and induces an sIBM-like pathology [7, 31]. DYSF is also reportedly increased in the endothelial cells of the blood–brain barrier and may serve as a marker of vessel damage that helps inflammatory cells pass through the vessel wall in multiple sclerosis [30]. These results indicate that DYSF may play a role in the process of inflammatory cell infiltration in the pathogenesis of DM and PM.

Here we analysed the global protein profiles of muscle tissues and found differentially and commonly altered proteins in DM and PM; DYSF was increased in both. Further, we investigated the molecular function of DYSF and explored the relationship between DYSF and HLA-A/MCP-1.

Methods

IIM patients and controls

Blood and muscle biopsies were collected from a total of 75 DM or PM patients enrolled at the Department of Rheumatology and Immunology of Xiangya Hospital of Central South University from June 2016 to May 2018 and 43 healthy controls (HCs). All patients fulfilled the 1975 Bohan and Peter criteria for a probable or definite DM/PM diagnosis. The exclusion criteria were infection, tumour or overlap syndrome, severe cardiovascular or metabolic disease, immunosuppressant use before admission and sIBM. All patients had proximal muscle weakness and showed myogenic damage on electromyography prior to enrolment. Forty-eight muscle biopsies were obtained from the patients' left biceps brachii muscle for diagnostic purposes and then immediately embedded in gum tragacanth or saved in liquid nitrogen (Supplementary Tables S1–S3, available at *Rheumatology* online). Twenty-eight control muscle biopsies were obtained from subjects undergoing shoulder replacement surgery. Serum samples were collected from 35 patients and 15 HCs (Supplementary Tables S4–S5, available at *Rheumatology* online). Twenty-nine whole blood samples were obtained from among the 75 patients and the specific samples used in each experiment are listed in Supplementary Table S6, available at *Rheumatology* online. Serum and embedded samples were stored at −80°C before use. Clinical data, including demographic characteristics, clinical manifestations, laboratory examinations and treatments were recorded. Myositis-specific and -related autoantibodies were detected using a commercial line blot (Euroimmun, Lübeck, Germany). The patients' response to treatment met at least one of two criteria: improved muscle weakness or serum creatinine kinase levels that normalized after therapy or decreased by >50% from baseline. The human experimentation protocol of this study was approved by the ethics committee of Xiangya Hospital (grant 201212074). The data were processed anonymously. All procedures in this study involving human participants were performed in accordance with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. All patients and HCs agreed to participate in the study and provided written informed consent.

Mass spectrometry and bioinformatics analyses

DM, PM and HC muscle tissue samples were analysed (Supplementary Table S1, available at *Rheumatology* online). Samples were sent to PTM BioLab, Hangzhou, China for processing. Tandem mass tag labelling (Thermo Fisher Scientific, Waltham, MA, USA) was performed according to the manufacturer's protocol. Expression data were analysed by *t*-test without any error correction for the samples. Proteins were considered statistically significant at $P < 0.05$. Cytoscape (https://cytoscape.org), ClueGO [33, 34] and the Database for Annotation, Visualization, and Integrated Discovery (DAVID; https://david.ncifcrf.gov/) version 6.8 [35, 36] were used to

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identify the networks, functions and canonical pathways of these proteins.

Flow cytometry

Flow cytometry was performed as described in previous reports [37, 38]. Briefly, 80 μL of fresh whole blood was obtained in EDTA tubes and immunolabelled with the PC5-conjugated anti-human CD56, PC7-conjugated anti-human CD14, APC-conjugated anti-human CD19, PE-conjugated anti-human CD33 (all from BD Biosciences, San Jose, CA, USA) and APC-Cy7-conjugated anti-human CD3 (Biolegend, San Diego, CA, USA) monoclonal antibodies. After lysis, permeabilization and washing, an anti-DYSF antibody (NCL-Hamlet; Leica Biosystems, Newcastle, UK) and FITC-conjugated goat anti-mouse IgG (H + L) (Jackson ImmunoResearch, West Grove, PA, USA) were used as secondary antibodies. The samples were then run on a flow cytometer (BD Biosciences) and analysed with FlowJo version X.0.7 software (FlowJo, Ashland, OR, USA).

Cell culture

Human skeletal muscle myoblasts (HSkMCs) from a single donor were cultured in Skeletal Muscle Cell Medium (ScienCell, San Diego, CA, USA). The HSkMCs were split at 60–65% confluency before the myoblasts touched each other. HSkMCs were induced to differentiate into myotubes by replacing the growth medium with a 1:1 mixture of DMEM-F12 medium (HyClone, Logan, UT, USA) containing 2% horse serum (HyClone) for 5–7 days after the cells reached 70% confluency. The cells were then cultured for an additional 8 days without subculturing.

Human dermal microvascular endothelial cells (HDMECs) were obtained from Cell Biolabs (#CBR130858; San Diego, CA, USA) and cultured in basic DMEM with 10% foetal calf serum (Hangzhou Sijiqing Biological Technology, Hangzhou, China). All cells were incubated at 37°C in a humidified atmosphere containing 5% carbon dioxide. IFN-β (PeproTech, Rocky Hill, NJ, USA) was added to the medium at concentrations of 0, 1, 10 and 100 ng/ml at 0, 6, 12, 24, 48 and 72 h and the cells were harvested in TRIzol reagent (Invitrogen Life Technologies, Waltham, MA, USA). Serum from patients and controls was added as described previously [17]. All experiments were repeated at least three times.

ELISA

DYSF concentrations in the serum of patients and controls were quantified using a human DYSF ELISA kit (Cusabio, Wuhan, China) according to the manufacturer's instructions. Briefly, 100 μL of serum or standard was incubated in the wells for 2 h at 37°C before removal. Biotin-conjugated antibody was added to each well and incubated for 1 h, followed by aspiration and three washes. Next, horseradish peroxidase–avidin was added and incubated for 1 h, followed by aspiration and five washes. 3,3′,5,5′-tetramethylbenzidine substrate was added to each well and incubated for 30 min in the dark. Stop solution was added and the absorbance was measured immediately at 450 nm with a microplate reader.

RNA extraction and real-time PCR

Total RNA from the cells and tissues was extracted using TRIzol according to the manufacturer's recommendations. RNA concentration and quality were measured by a QS3000 spectrophotometer (Quawell, San Jose, CA, USA). One microgram of RNA was reverse transcribed using the reverse transcription system (Promega, Madison, WI, USA). Quantitative real-time PCR was performed using TB Green PCR Master Mix (Takara, Shiga, Japan) on a 7500 Real-Time PCR system (Applied Biosystems, Waltham, MA, USA). The relative expression of target genes was calculated by the 2-ΔΔCT method and normalized to glyceraldehyde 3-phosphate dehydrogenase (GAPDH). The details of the gene-specific primers are shown in Supplementary Table S7, available at *Rheumatology* online.

Co-immunoprecipitation and Western blot analysis

For the immunoprecipitation assays, protein extracts from human muscle tissues were incubated with a mouse anti-human DYSF antibody (1:25; NCL-Hamlet), rabbit anti-human HLA-A antibody (1:25; Abcam, Cambridge, UK) and Protein A + G Sepharose Beads (Millipore, Burlington, MA, USA) for 16 h at 4°C. The next day the beads were washed extensively and prepared for Western blot analysis. In the protein analysis, 60 μg protein aliquots were separated by 8% SDS-PAGE and transferred onto polyvinylidene difluoride membranes. Thereafter the membranes were blocked with 5% non-fat dried milk in Tris-buffered saline with 0.1% Tween for 2 h and incubated with anti-GAPDH (1:2000), anti-DYSF (1:500) or anti-HLA-A (1:2000) antibodies at 4°C overnight. The next day, the membranes were washed and incubated with secondary antibodies. The enhanced chemiluminescence method (Bio-Rad, Hercules, CA, USA) was used to quantify the Western blot results. Co-immunoprecipitation was repeated three times for the DM and PM patients.

IF staining

IF staining of the tissues was performed on frozen 10 μm sections. Neutrophils and peripheral blood mononuclear cells were isolated by density gradient equilibrium centrifugation using Ficoll 1077 and 1119 (Sigma-Aldrich, St. Louis, MO, USA) and then mounted onto slides. All slides were fixed with 4% paraformaldehyde or acetone/methanol (1:1) for the DYSF (NCL-Hamlet) antibody for 15 min. After three washes, permeabilization with 0.1% Triton X-100 in PBS for 15 min and blocking with 5% goat serum in PBS for 30 min were performed and the slides were incubated overnight at 4°C with mouse anti-human DYSF (1:25; NCL-Hamlet), rabbit anti-human DYSF, rabbit anti-human HLA-A, mouse anti-human CD68, rabbit anti-human CD15, mouse anti-human MCP-1, rabbit anti-human CD4, rabbit anti-human CD8, rabbit anti-human CD14, rabbit anti-human CD19

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(all used at 1:100; Abcam) and mouse anti-human CD56 (1:15; Santa Cruz Biotechnology, Dallas, TX, USA) antibodies. The corresponding secondary antibodies were incubated for 1 h at room temperature in the dark and anti-fading mounting medium with 4′,6-diamidino-2-phenylindole (Solarbio, Beijing, China) was then added. Unrelated isotype-matched antibodies were used as controls. Images were captured using an Olympus microscope with a DP72 camera (Olympus, Shinjuku, Japan). Linear adjustments of the images were made using Photoshop CS5 (Adobe, San Jose, CA, USA) or ImageJ version 1.51 (National Institutes of Health, Bethesda, MD, USA) software. Each staining experiment was repeated three times with samples from different patients and representative staining samples are shown.

Terminal deoxynucleotidyl transferase nick end labelling analysis

Apoptotic cells in muscle tissue and among HSkMCs grown on coverslips stimulated with 60 μmol/ml hydrogen peroxide (H_2O_2) at 2 h were double-labelled by terminal deoxynucleotidyl transferase 2′-deoxyuridine 5′-triphosphate nick end labelling (TUNEL; TMD Red In Situ Cell Death Detection Kit; Roche Applied Science, Mannheim, Germany) and DYSF according to the manufacturer's instructions. In brief, the slides were air dried, fixed in 4% paraformaldehyde in PBS at room temperature for 20 min, permeabilized with 0.1% Triton X-100 in 0.1% sodium citrate on ice for 2 min and incubated in a TUNEL reaction mixture in a humidified chamber at 37°C for 1 h in the dark. Negative control experiments were performed in which the terminal deoxynucleotidyl transferase enzyme was not added to the TUNEL reaction mixture. After three washes, all slides were subjected to IF staining.

Statistical analysis

All data were analysed using SPSS version 18.0 for Windows (SPSS, Chicago, IL, USA) and statistical significance was set as a two-sided P-value <0.05. Quantitative variables are described as mean (s.d.) for normally distributed data and as median (interquartile range) for skewed data by the Wilcoxon rank sum test. Categorical variables are presented as numbers and analysed by the χ^2 test or Fisher's exact test. Correlations between gene expression values and clinical examinations were analysed by Spearman's correlation coefficient.

Results

Mass spectrometry analysis of muscle tissue in DM and PM

We performed a mass spectrometry analysis of three DM, three PM and 3 HC samples and identified a total of 2942 proteins. By comparing the protein levels with those in HC samples, we identified 125 upregulated and 37 downregulated proteins in DM and 135 upregulated and 82 downregulated proteins in PM ($P < 0.05$). Among these, 27 genes were commonly upregulated and 25 genes were commonly downregulated in the DM and PM samples (Fig. 1A–D, I). GO annotation of protein function revealed an enriched purine nucleoside monophosphate metabolic process, muscle contraction, a coenzyme metabolic process and generation of precursor metabolites and energy and protein targeting to the membrane in DM samples (Fig. 1E), while the regulation of calcium ion transmembrane transporter activity, an aromatic compound catabolic process, fatty acid oxidation, energy derivation by oxidation of organic compounds and mitochondrial transmembrane transport were enriched in PM samples (Fig. 1G). A Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis of the upregulated proteins suggested that glycolysis/gluconeogenesis; the pentose phosphate pathway; fructose and mannose metabolism; galactose metabolism and alanine, aspartate and glutamate metabolism were enriched in DM samples (Fig. 1F). Ribosome, antigen processing and presentation and protein processing in the endoplasmic reticulum, proteasome and spliceosome were enriched in PM samples (Fig. 1H).

According to the global protein analysis of muscle tissues, we found that DM and PM have many shared differentially expressed proteins as well as specific proteins. A bioinformatics analysis suggested that the dysregulated proteins in DM and PM samples participated in common biological processes and pathways, such as the generation of precursor metabolites and energy, glycolysis/gluconeogenesis and the HIF-1 signalling pathway, but also participated in specific pathways such as the pentose phosphate pathway in DM samples and antigen processing and presentation in PM samples.

High DYSF expression in DM/PM correlated with clinical features

The most commonly altered proteins were further validated in 42 patients and 28 HCs. Real-time PCR confirmed that DYSF, SMTNL2, HMGB1, ART3, SND1 and CRAT were significantly increased in the muscle tissues of DM and PM patients (Fig. 2A). IF of muscle tissue sections showed that in HC samples, DYSF was expressed mainly on the plasma membrane of muscle fibres and weakly in the cytoplasm of muscle fibres or in the perimysium. However, the DYSF level was increased in the plasma membrane and cytoplasm in the muscle fibres of DM patients, especially those with perifascicular atrophy myofibres. DYSF was also elevated in the plasma membrane and the cytoplasm of scattered myofibres with inflammatory cell infiltration into the endomysium in PM samples (Fig. 2B). We then determined the concentration of DYSF in the serum and found that the levels of DYSF were elevated in DM and PM samples (Fig. 2C). Although the levels of DYSF displayed different patterns in patients with specific antibodies, no statistically significant differences were observed (Fig. 2D). Glucocorticoid dose was not associated with the DYSF level (Fig. 2E). DYSF levels were correlated with CRP ($r = 0.334$, $P = 0.049$) (Fig. 2I) but not myocardial enzymes ($r = 0.012$, $P = 0.534$) (Fig. 2F).

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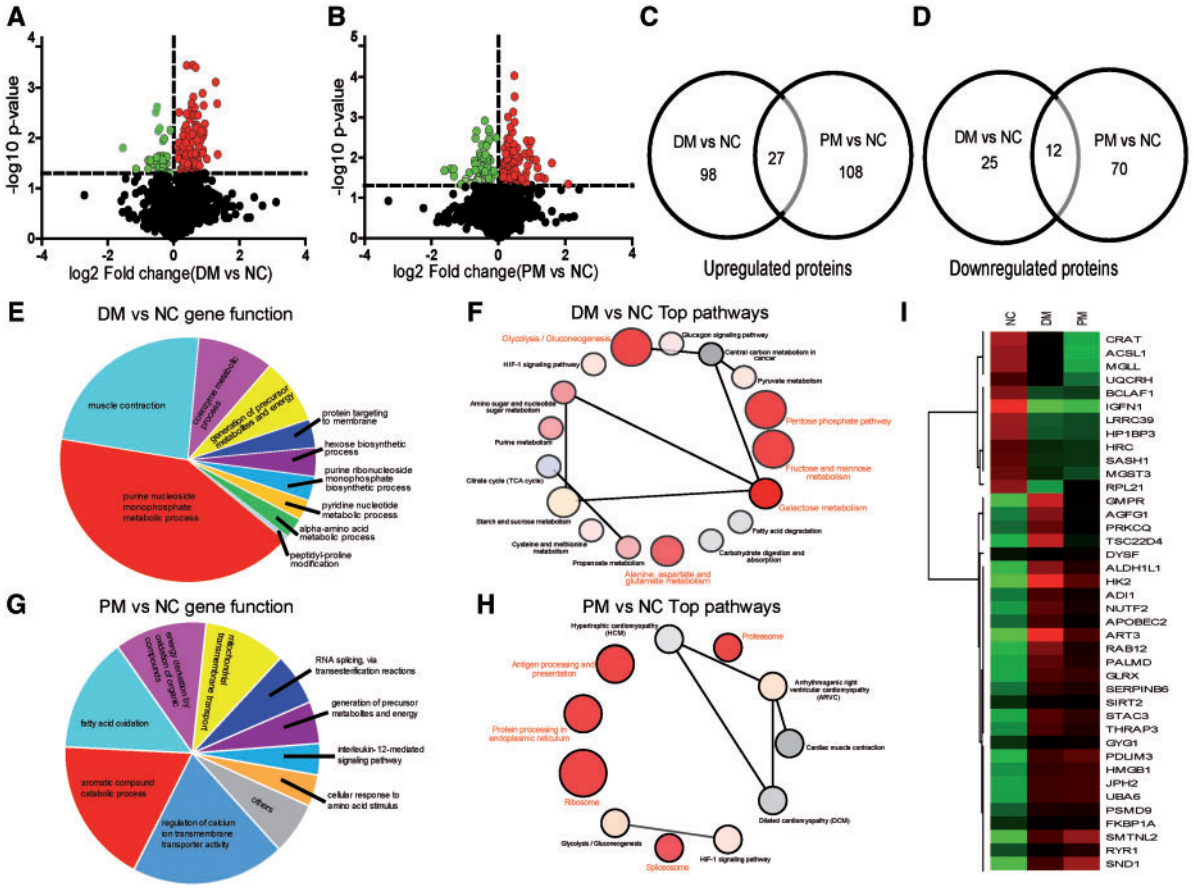
Fig. 1 Mass spectrometry and bioinformatics analysis of muscle tissues in DM/PM



(A, B) Differentially expressed proteins in the muscle tissues of DM and PM patients compared with HCs. **(C, D)** Venn diagram shows the number of upregulated and downregulated proteins in DM and PM. Twenty-seven upregulated proteins and 12 downregulated proteins were altered in both DM and PM. **(E, G)** GO annotation of the dysregulated proteins in DM and PM. **(F, H)** KEGG pathway analysis suggests the top pathways in DM and PM. **(I)** Heat map shows the commonly up- and downregulated proteins in DM and PM.

DYSF co-located with HLA-A in muscle tissue

MHC-I molecules are not constitutively displayed in normal muscle tissue but are seen in some intermuscular capillary endothelial cells. We found that three types of MHC-I molecules (HLA-A, HLA-B and HLA-C) were increased in DM and PM and correlated with DYSF (Fig. 3A). However, only unique peptides of HLA-A were identified by mass spectrometry. Intriguingly, high levels of DYSF and HLA-A proteins were found in DM and PM muscles and a positive correlation was observed (Fig. 3B). These results were confirmed by IF staining of muscle tissues (Fig. 3C). Moreover, we found that a large number of myofibres in DM and PM samples that expressed DYSF were co-located with HLA-A on the cytomembrane and in the cytoplasm (Fig. 3C). Dysferlinopathy patients with very little DYSF expression on the membrane also had faint HLA-A expression in myofibres (Fig. 3C). Patients with other progressive muscular dystrophy diseases, such as Duchenne muscular dystrophy and facioscapulohumeral muscular dystrophy, have much lower DYSF and HLA-A expressions than DM and PM patients (Fig. 3C). Similar to myofibres, the co-localization of DYSF and HLA-A in the perimysium with enlarged vessels and inflammatory cell infiltration into muscles was also observed in DM and PM samples (Fig. 3C).

DYSF expression positively correlated with HLA-A

IFN-β is a pro-inflammatory cytokine that reportedly induces MHC-I overexpression [39–41]. To explore the relationship between DYSF and HLA-A, we used 20% DM/PM patients' serum or IFN-β to stimulate muscle cells (myoblasts and myotubes) and endothelial cells. After stimulation, DYSF and HLA-A expressions were increased in a time- and dose-dependent manner and showed a similar tendency (Fig. 4A–D). However, we failed to find a direct interaction between DYSF and HLA-A using co-immunoprecipitation (Fig. 4E).

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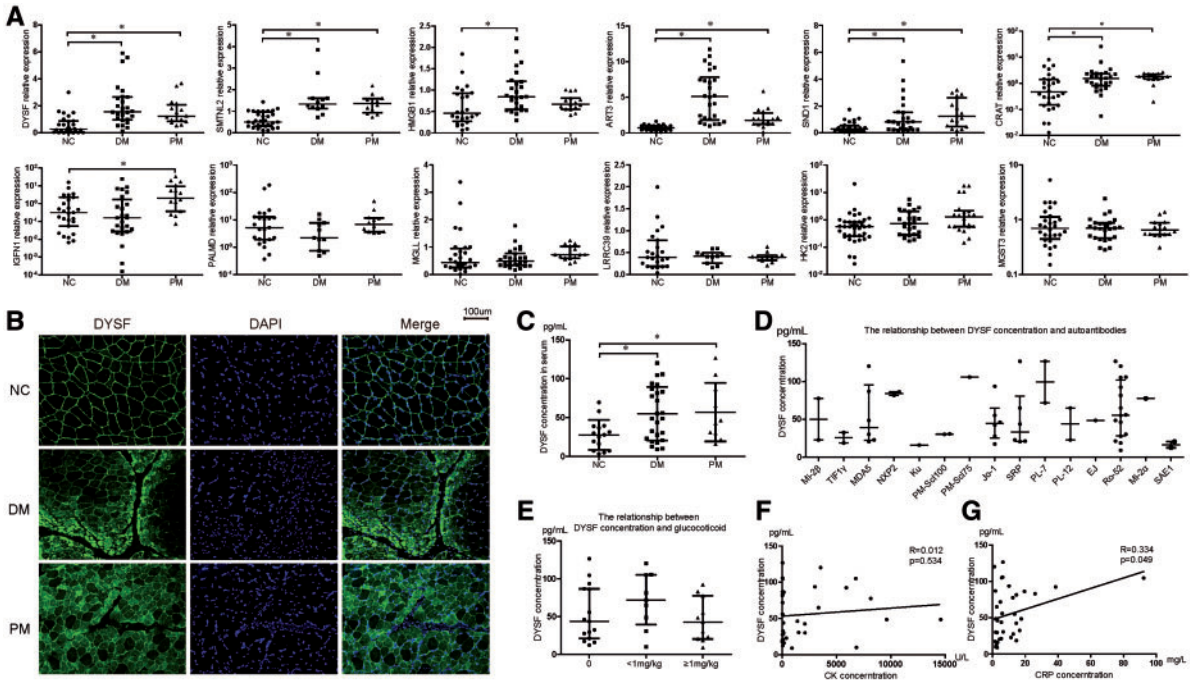
Fig. 2 DYSF was elevated in muscle tissue of DM and PM and correlated with clinical features



(A) mRNA expression level of selected proteins that upregulated in DM and PM muscle tissue. **(B)** IF of muscle tissue among DM (patient 16), PM (patient 39) and HCs. DYSF only expressed in sarcolemma of muscle fibres in HCs. In DM and PM, DYSF was expressed in both sarcolemma and cytoplasm. **(C–G)** DYSF concentrations were increased in the serum of DM/PM and correlated with CRP levels but was not correlated with myositis antibodies, glucocorticoid dose or creatinine kinase levels. $^{*}P < 0.05$ compared with HCs.

To determine whether increased DYSF was involved in myofibre apoptosis and regeneration, we used double staining for TUNEL and DYSF in muscle tissues and H_2O_2-induced myoblasts *in vitro*. We found no change in DYSF expression during apoptosis in myofibres (Supplementary Figs. S1A, 1B, available at *Rheumatology* online). Previous studies reported that DYSF may have an important role in myocyte regeneration. However, double immunostaining of neonatal muscle heavy chains and DYSF showed that only a few regenerated muscle cells strongly expressed DYSF (Supplementary Fig. S1C, available at *Rheumatology* online).

DYSF expression in inflammatory cells from DM and PM patients

Since muscle tissues from DM/PM patients are infiltrated with inflammatory cells [38], we also investigated DYSF expression in the peripheral blood and explored the potential relationship between DYSF and inflammatory cells. In the peripheral blood, only $CD14^+$ monocytes and neutrophils showed large amounts of DYSF in the cytoplasm, especially near the nucleus. $CD4^+$ T cells, $CD8^+$ T cells, $CD19^+$ B cells and $CD56^+$ NK cells had barely detectable levels of DYSF. However, no statistically significant differences were observed among the DM, PM and HC samples by IF and flow cytometry (Fig. 5A, B). Regarding the high level of DYSF in neutrophils and monocytes, DYSF was co-stained in the muscle tissue. Compared with that of abnormal myofibres, $CD15^+$ neutrophils and $CD68^+$ macrophages showed lower DYSF expression in DM and PM samples (Fig. 5C, E). However, it should be noted that $CD68^+$ macrophages were mainly distributed in the perimysial areas near myofibres with perifascicular atrophy in DM samples and among scattered myofibres in PM samples and strongly expressed DYSF (Fig. 5C). Furthermore, in DM and PM patients with $CD68^+$ macrophage infiltration, DYSF was largely co-located with MCP-1 (Fig. 5D).

Discussion

The current study is the first to demonstrate that DYSF is increased in DM and PM muscle tissues and might be involved in the pathogenesis of DM and PM. Elevated DYSF was mainly expressed in the cell membrane and cytoplasm of injured myofibres in DM and PM samples. Moreover, high levels of DYSF were mainly located in DM samples with perifascicular atrophy and in scattered areas of PM samples with inflammatory cell infiltration. Glover *et al.* [42] determined that overexpressed DYSF in a mouse model induced a striking phenotype of reduced muscle mass and strength and that its level was

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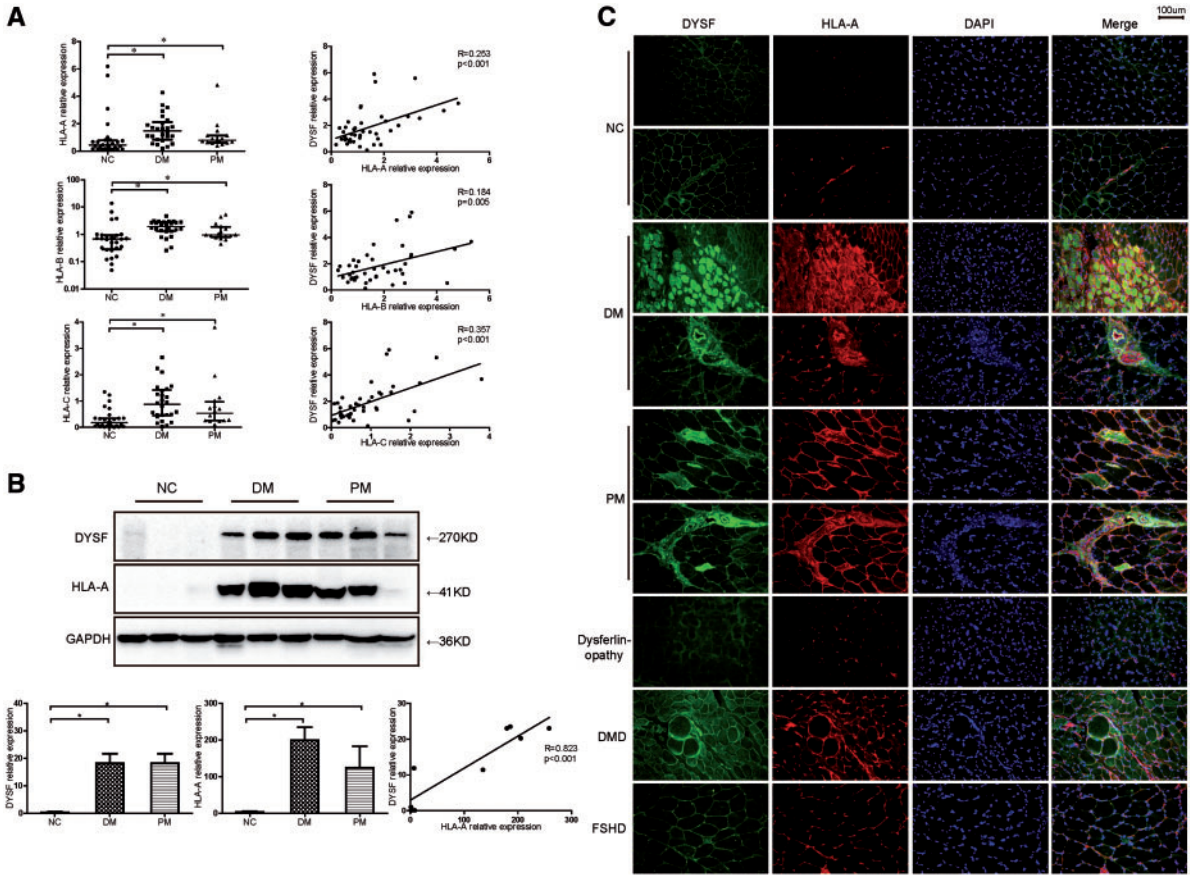
Fig. 3 DYSF co-located with HLA-A in muscle tissues of DM/PM



(A) mRNA expression of three classic MHC-I molecules (HLA-A, HLA-B and HLA-C) was elevated and correlated with DYSF expression in muscle tissue of DM and PM. **(B)** Western blot of muscle tissue lysis from DM, PM and HCs. To a great extent DYSF paralleled the HLA-A standard at the protein level (patients' samples from left to right: 24, 25, 26, 46, 36 and 41). **(C)** Co-staining of DYSF and HLA-A in muscle tissues of DM (patient 12), PM (patient 7) and HCs. Dysferlinopathy, DMD and FSHD served as disease controls and displayed a different pattern compared with DM and PM. $^*P < 0.05$ compared with HCs. DMD: Duchenne muscular dystrophy; FSHD: facioscapulohumeral muscular dystrophy.

significantly correlated with muscle weakness severity. Considering previous studies in combination with our results, we speculated that a high level of DYSF might be associated with inflammatory cell infiltration and muscle damage. Progressive muscle weakness and muscle enzyme elevation are major clinical manifestations in DM, PM, dysferlinopathy, facioscapulohumeral muscular dystrophy and Duchenne muscular dystrophy. However, specific markers for distinguishing among these disorders have yet to be identified [32, 43, 44]. Here we found that DYSF was specifically highly expressed in DM and PM samples, suggesting its role as a potential biomarker for DM and PM.

We also found that serum DYSF levels were relatively high in DM/PM patients and correlated with CRP levels. Dong *et al.* [45] determined that DYSF can exist in exosomes and be transferred to other myotubes and improve the membrane repair capabilities of human $DYSF^{-/-}$ myotubes irrespective of mutations. However, the function of serum DYSF remains unknown. As DYSF is highly expressed in myofibres and endothelial cells in the muscle tissues of DM/PM patients, we speculate that serum DYSF is mainly formed by necrotic myofibres, which may reflect the extent of muscle damage. Also, we found that DYSF mainly exists in monocytes and neutrophils in the peripheral blood and may interact with adhesion proteins that regulate leucocyte migration to specific tissues [46]. However, DYSF expression in monocytes and neutrophils was not elevated in DM and PM patients in our study. These results also indicate that the high expression of DYSF in DM/PM muscle tissues was not caused by inflammatory cells.

T cell and macrophage infiltration into muscle tissue and subsequent damage to myofibres are critical in DM/PM [47]. MHC-I can be recognized by $CD8^+$ cytotoxic T lymphocytes and mediate target cell damage by the

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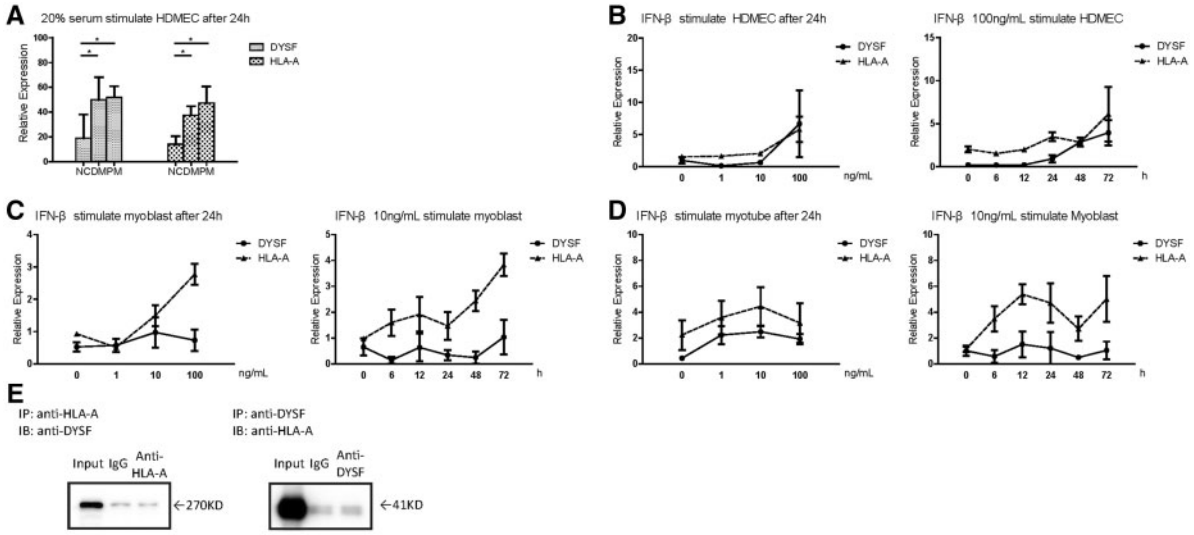
Fig. 4 DYSF expression correlated with HLA-A in muscle cells and endothelial cells *in vitro*



(A) The expression of DYSF and HLA-A in HDMEC after treating with 20% serum from DM, PM and HCs for 24 h. **(B–D)** The expression of DYSF and HLA-A in myoblasts, myotubes and HDMEC after treating with 0, 1, 10 and 100 ng/ml IFN-β for 24 h. Both DYSF and HLA-A in myoblasts, myotubes and HDMEC were increased. The expression of DYSF and HLA-A at 0, 6, 12, 24, 48 and 72 h after IFN-β stimulation was also detected. A similar trend was observed between DYSF and HLA-A. **(E)** Co-immunoprecipitation of DYSF and HLA-A in muscle biopsies of DM/PM patients (patients 28 and 39). $^{*}P < 0.05$ compared with HCs.

perforin–granzyme B and Fas–FasL pathways [48–50]. MHC-I is considered a key molecule that facilitates inflammatory cell infiltration, leading to muscle damage and its wide use as a biomarker in the diagnosis of DM/PM [51]. HLA-A, the most highly expressed MHC-I molecule, is closely related to the pathogenesis of DM/PM [52, 53]. In our study, both DYSF and HLA-A were upregulated and co-located in the cell membrane and cytoplasm of myofibres, and very low expression of DYSF and HLA-A was found in dysferlinopathy patients [54, 55]. These studies suggest that DYSF is closely related to HLA-A, which might participate in the pathogenesis of inflammatory cell infiltrations in IIM. Tapasin and TAP-binding protein-related are IFN-inducible transmembrane glycoproteins that can interact with HLA-A to regulate its transport across the endoplasmic reticulum membrane [56]. DYSF is also an IFN-inducible protein that lies on the membrane of vesicles and sarcolemma and induces cytokine-filled vesicles to fuse with the sarcolemma in a Ca^{2+}-dependent manner [57, 58]. Therefore DYSF may have a role in HLA-A transportation to the cell membrane and secretion into the extracellular space. However, we did not observe that DYSF and HLA-A interact directly in the muscle tissue. DYSF may be an upstream or downstream molecule in the pathway by which IFN-β upregulates HLA-A expression in IIM. The exact mechanism of this phenomenon requires further study.

By expressing MHC-I on the cell surface to activate T cells, monocytes can secrete chemokines to attract circulating white blood cells under inflammatory conditions [59]. Miyatake *et al.* [60] found that muscle cells can release MCP-1 through the nuclear factor κB–regulated response to promote monocyte infiltration into the skeletal muscle. DYSF-deficient mice showed reduced inflammatory cell recruitment and DYSF-deficient primary myoblasts showed impaired MCP-1 secretion upon IFN-β stimulation [57]. These results indicated that DYSF facilitates MCP-1 release, resulting in monocyte infiltration. We found that DYSF and MCP-1 co-localized in impaired muscle fibres of patients with IIM, especially those with perifascicular atrophy. Moreover, macrophages infiltrated almost exclusively around the myofibres with increased DYSF in IIM. The findings also suggest that increased DYSF expression is involved in the infiltration of monocytes during the development of muscle damage in IIM.

This study has some limitations. We demonstrated that DYSF is associated with two important molecules, HLA-A and MCP-1, in IIM; however, the exact mechanism of DYSF in IIM remains unknown. According to the autoantibody and pathology findings, the patients in our study had different clinical phenotypes, but we did not investigate their individual levels. Additionally, we did not obtain muscle sections and serum samples from the same patients or include sIBM and immune-mediated necrotizing myopathy patients. Therefore it was difficult to assess the potential clinical use of DYSF. Further studies are needed to confirm the role of DYSF in the pathogenesis of IIM.

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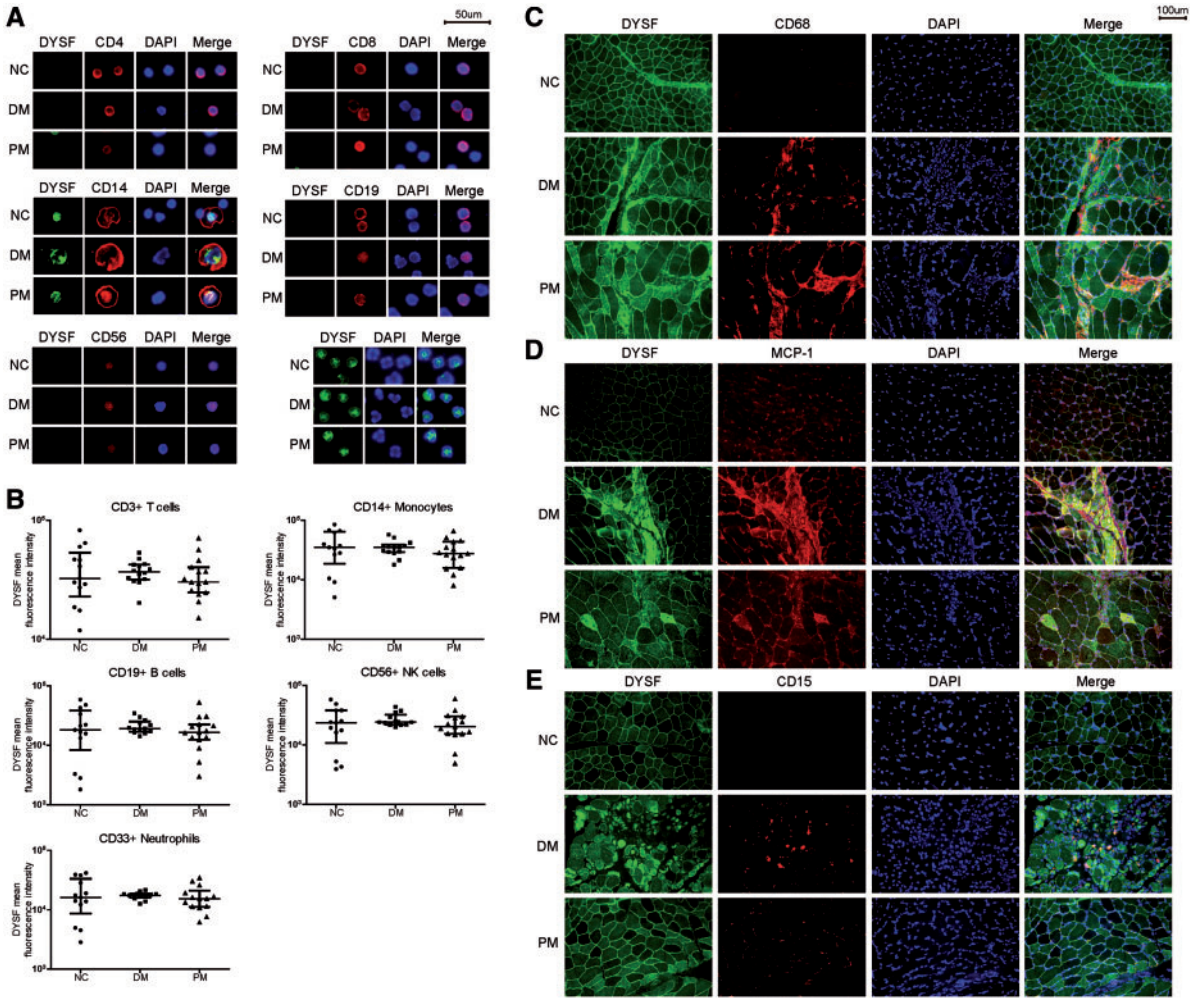
Fig. 5 DYSF in inflammatory cells and co-located with MCP-1 in muscle tissue



(A) IF of DYSF in CD4$^+$ T cells, CD8$^+$ T cells, CD14$^+$ monocytes, CD19$^+$ B cells, CD56$^+$ NK cells and neutrophils from peripheral blood in DM (patient 9), PM (patient 41) and HCs. **(B)** Mean fluorescence intensity of DYSF in CD3$^+$ T cells, CD14$^+$ monocytes, CD19$^+$ B cells, CD56$^+$ NK cells and CD33$^+$ neutrophils by flow cytometry. **(C, E)** Co-staining of DYSF with CD68$^+$ macrophages and CD15$^+$ neutrophils in muscle tissue of DM, PM and HCs. DM (patient 13) and PM (patient 45) are shown in C and DM (patient 10) and PM (patient 41) are shown in E. **(D)** Double IF of DYSF and MCP-1 in muscle tissue of DM, PM and HCs. DYSF co-located with MCP-1 in myofibres of DM (patient 13) and PM (patient 38).

Conclusion

In conclusion, high levels of DYSF were found in the myofibres and endothelium of muscle tissue as well as the serum in DM and PM. Increased DYSF levels in muscle tissue are associated with two important molecules, MHC-I and MCP-1, which may be involved in the process of inflammatory cell infiltration and muscle cell damage. Therefore DYSF may play an important role in the development and progression of DM and PM.

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Supplementary data

Supplementary data are available at *Rheumatology* online.

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